

A Fourier-Positive Even-Exponent Subcase of the Reverse Bernstein Problem on the Hypercube

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Status: candidate partial result, likely valid. The general scalar reverse Bernstein problem for arbitrary Fourier signs and arbitrary $p \in (1, \infty) \setminus \{2\}$ remains open. This packet proves the sharp estimate for every even exponent when all Walsh coefficients can be made nonnegative by a translation of the cube.

1 Source and original problem

The source is P. Ivanisvili, R. Vershynin, and X. Xie, *Jackson's inequality on the hypercube*, arXiv:2410.19949v1 [1]. The problem appears in Section 1.3, page 3 of the arXiv PDF (source file `source.tex`, line 314). In the source normalization,

$$\Delta f(x) = \sum_{j=1}^n (f(x) - f(x^j)),$$

where x^j is obtained by flipping coordinate j . Thus $\Delta W_S = 2|S|W_S$ for the Walsh character

$$W_S(x) = (-1)^{\sum_{j \in S} x_j}.$$

The tail space $L_{\geq d}^p$ consists of the functions whose Walsh coefficients vanish on all levels $|S| < d$.

The source states that, for $p \in (1, \infty)$, the reverse Bernstein estimate

$$\|f\|_p \leq \frac{C_p}{d} \|\Delta f\|_p, \quad f \in L_{\geq d}^p, \quad (1)$$

remains a major open problem. The case $p = 2$ is immediate from Parseval, with $C_2 = 1/2$, so the substantive open range is $p \neq 2$.

2 Definitions and scope of the partial result

For $y \in \{0, 1\}^n$, let $\tau_y f(x) = f(x \oplus y)$, where $\oplus$ denotes coordinatewise addition modulo two. Since

$$\widehat{\tau_y f}(S) = W_S(y) \widehat{f}(S),$$

we make the following definition.

Fourier-positive up to translation. A real-valued $f : \{0, 1\}^n \rightarrow \mathbb{R}$ has this property if there is a $y \in \{0, 1\}^n$ such that

$$W_S(y) \widehat{f}(S) \geq 0 \quad \text{for every } S \subseteq [n].$$

Equivalently, one translate of f has nonnegative Walsh coefficients.

This is a genuine sign restriction. It is preserved by multiplying f by a positive scalar and includes single Walsh characters after a translation and choice of positive coefficient. It does not include general tail functions.

1.3. Failure of reverse Bernstein inequality in the tail space $L_{\geq 0.499n}^1$. For any $d \in [0, n]$ and any $p \geq 1$, let $L_{\geq d}^p$ denote the d 'th tail space equipped with the norm $\|\cdot\|_p$ and consisting of all functions f on the hypercube that can be represented as $f(x) = \sum_{|S| \geq d} a_S W_S(x)$. Set $\Delta f(x) = \sum_{j=1}^n (f(x) - f(x^j))$. Theorem 1 gives the following

Corollary 3. *For any $c_1 \in (0, 1/2)$, there exists $c_2 > 0$ that depends only on c_1 and such that for any $n \geq 1$ there exists a function $f \in L_{\geq c_1 n}^1$, $f \neq 0$, with the property*

$$\|f\|_1 \geq c_2 \|\Delta f\|_1.$$

In [MN14] Mendel and Naor showed that if the reverse Bernstein inequality $\|f\|_p \leq \frac{C_p}{n} \|\Delta f\|_p$ were to hold in the tail space $L_{\geq n/10}^p$ for vector-valued functions then this would simplify their construction of vector-valued expanders (see Remark 7.5 in [MN14]). They also proved (see [MN14, equation 145]) that the inequality fails in the tail space $L_{\geq c \log \log n}^1$. Corollary 3 shows that one can find a counterexample to the reverse Bernstein inequality even in the smaller space $L_{\geq cn}^1$. Before we conclude this section, we point out that for any $p \in (1, \infty)$, the reverse Bernstein inequality in the tail space $L_{\geq d}^p$ remains a major open problem, see [HMO17, EI20, EI23] for some partial results.

Figure 1: Full-width crop of Section 1.3 on page 3 of the source paper.

3 Idea of the proof

For an even exponent $p = 2m$, both $\|f\|_p^p$ and the dual pairing $\mathbb{E}[f^{p-1} \Delta f]$ can be expanded as sums over $2m$ Walsh frequencies whose symmetric difference is empty. When the Walsh coefficients are nonnegative, every summand is nonnegative. Symmetry among the $2m$ frequencies lets us replace the Laplacian weight on the last frequency by the average of all $2m$ weights. Every active frequency has size at least d , so that average is at least $2d$ in the source normalization. Holder's inequality then converts the pairing estimate into the desired norm estimate. A single level- d character shows that the constant $2d$ is sharp.

The positivity is precisely what prevents cancellations in the multilinear Walsh expansion. Removing it is the unresolved obstacle; the argument does not claim progress for arbitrary Fourier signs.

4 The partial theorem

Theorem 1 (Sharp Fourier-positive even- p reverse Bernstein estimate). *Let $n, d, m \in \mathbb{N}$ with $1 \leq d \leq n$, and put $p = 2m$. Suppose that $f : \{0, 1\}^n \rightarrow \mathbb{R}$ satisfies*

$$\widehat{f}(S) = 0 \quad \text{when } |S| < d$$

and is Fourier-positive up to translation. Then

$$\|\Delta f\|_p \geq 2d \|f\|_p. \tag{2}$$

The factor $2d$ is optimal for every even p .

Proof. Translations commute with Δ and preserve every L^p norm. We may therefore translate f and assume that

$$a_S := \widehat{f}(S) \geq 0 \quad \text{for every } S \subseteq [n].$$

The tail hypothesis says that $a_S = 0$ whenever $|S| < d$.

Write $p = 2m$. Orthogonality of Walsh characters gives

$$\mathbb{E} f^{2m} = \sum_{S_1 \Delta \dots \Delta S_{2m} = \emptyset} \prod_{j=1}^{2m} a_{S_j}. \tag{3}$$

Indeed, the expectation of $W_{S_1} \cdots W_{S_{2m}}$ is one exactly when $S_1 \triangle \cdots \triangle S_{2m} = \emptyset$, and is zero otherwise.

Since $\Delta W_S = 2|S|W_S$, the same expansion yields

$$\mathbb{E}[f^{2m-1} \Delta f] = \sum_{S_1 \triangle \cdots \triangle S_{2m} = \emptyset} 2|S_{2m}| \prod_{j=1}^{2m} a_{S_j}. \quad (4)$$

The indexing condition and the coefficient product are symmetric in $S_1, \dots, S_{2m}$. Consequently, the value of the sum in (4) is unchanged if $|S_{2m}|$ is replaced by $|S_r|$ for any r , and hence

$$\mathbb{E}[f^{2m-1} \Delta f] = \frac{1}{2m} \sum_{S_1 \triangle \cdots \triangle S_{2m} = \emptyset} 2 \left(\sum_{j=1}^{2m} |S_j| \right) \prod_{j=1}^{2m} a_{S_j}. \quad (5)$$

Every product in (5) is nonnegative. If it is nonzero, then $a_{S_j} > 0$ for every j , so the tail hypothesis gives $|S_j| \geq d$ for every j . Therefore

$$\frac{2}{2m} \sum_{j=1}^{2m} |S_j| \geq 2d$$

term by term. Comparing (5) with (3), we obtain

$$\mathbb{E}[f^{p-1} \Delta f] \geq 2d \mathbb{E} f^p = 2d \|f\|_p^p, \quad (6)$$

where the last equality uses that p is even.

Holder's inequality and (6) now imply

$$2d \|f\|_p^p \leq \mathbb{E}[f^{p-1} \Delta f] \leq \|f^{p-1}\|_{p/(p-1)} \|\Delta f\|_p = \|f\|_p^{p-1} \|\Delta f\|_p.$$

If $f \neq 0$, division by $\|f\|_p^{p-1}$ proves (2); the zero case is trivial.

Finally, take $f = W_S$ with $|S| = d$. Then $|f| = 1$ and $\Delta f = 2df$, so equality holds in (2). This proves sharpness. $\square$

Remark 1 (Alternative Laplacian normalization). *Some cited papers use $L = \Delta/2$, so that $LW_S = |S|W_S$. In that normalization Theorem 1 reads $\|Lf\|_p \geq d\|f\|_p$.*

5 Adversarial verification

Step check

Step	Verdict	Check
Translation reduction	Valid	τ_y is a cube automorphism, commutes with Δ , preserves the uniform measure, and changes coefficient signs by $W_S(y)$.
Walsh moment expansion	Valid	The product of characters is the character indexed by the symmetric difference; its mean is zero unless that index is empty.
Termwise positivity	Valid	This is exactly where nonnegative translated Fourier coefficients are used. Zero coefficients cause no difficulty.
Symmetrization of weights	Valid	The constraint and coefficient product are invariant under all permutations of the $2m$ frequency indices.
Tail lower bound	Valid	A nonzero product forces all $2m$ participating coefficients to be nonzero, hence all frequency sizes are at least d .
Holder and sharpness	Valid	The dual exponent is $p/(p-1)$, and a single level- d character gives equality.

Finite checks

The reusable script `code/verify_fourier_positive_even_p.py` generated independent exponential nonnegative coefficients for every $2 \leq n \leq 6$, every $1 \leq d \leq n$, and $p \in \{4, 6, 8\}$. Ten trials per parameter triple gave 600 checks and no violation. The smallest reported slack in $\|\Delta f\|_p / \|f\|_p - 2d$ was -1.78×10^{-15} , numerical roundoff at the equality case $d = n$. The command was

```
conda run --no-capture-output -n sandbox python \
  code/verify_fourier_positive_even_p.py --max-n 6 --trials 10
```

These checks are not used in the proof.

The separate script `code/finite_tail_inverse_probe.py` performs a nonconvex numerical search without the sign restriction. For $n \leq 6$ and d near $n/2$, probes at $p = 1.5$ and $p = 4$ kept $d\|\Delta^{-1}\|$ near 0.51–0.55, compared with the exact Hilbert value $1/2$. This found no small counterexample to the full conjecture, but it cannot certify an operator norm or prove the conjecture.

Verifier verdict

Likely valid partial result. The proof is finite, algebraic, and has no external mathematical dependency beyond elementary Walsh orthogonality and Holder’s inequality. Human review should focus on the permutation averaging in (5) and the factor two coming from the source paper’s definition of Δ .

6 Failed full-solution routes and remaining obstruction

Three broader routes were tested before isolating Theorem 1.

1. Integrating the best known pure- L^p heat-smoothing estimate gives only a power $d^{-\alpha_p}$ with $\alpha_p < 1$ for $p \neq 2$, not d^{-1} .
2. The nonlinear L^p Poincare pairing controls mean-zero functions but does not see the tail depth d ; inserting it loses the required factor.
3. A tensor-product construction does not immediately yield a counterexample because the tilted mean of the block logarithmic Laplacian is positive. Making that mean small is essentially another form of the open inequality.

For arbitrary Fourier signs, the sums in (3) and (4) contain cancellations. The weighted sum need not dominate the unweighted moment term by term. No argument in this packet controls those cancellations, and no claim is made for non-even p .

7 Novelty and literature-status check

The bounded check was performed on 19 July 2026 after the direct proof attempt. It covered the run’s four lightweight indexes, the 35,297-paper local source-probed corpus, exact-phrase searches for “reverse Bernstein” and “tail space”, searches for the heat-smoothing conjecture, and searches for the combinations “nonnegative Fourier coefficients”, “Fourier-positive”, “Hamming cube”, and “Laplacian”. The close papers found were:

- Eskenazis–Ivanisvili [2], which proves sublinear-exponent bounds in general and a sharp-order result for narrow spectra;
- later work on complex hypercontractivity and the heat-smoothing exponent, including arXiv:1907.11359;
- Domelevo et al. [3], which proves linear heat decay with mixed L^2 and neighboring- L^p norms, not the pure estimate (1).

No exact match for Theorem 1 was found. This bounded search does not establish novelty, so novelty confidence is *moderate*. The theorem should be treated as a candidate new partial result until a specialist checks the Fourier-analytic literature.

8 Limitations and human-review recommendation

Theorem 1 settles an infinite, sharp subcase of the scalar question, but it does not settle the general problem. Its two restrictions are substantial:

- p must be an even positive integer;
- after a cube translation, all Walsh coefficients must be nonnegative.

The recommended human action is to verify the four-line multilinear expansion and then assess whether the positivity mechanism is known under a different name (for example, positive-definite functions on the Boolean group). A secondary research direction is to replace coefficientwise positivity by a weaker $2m$ -fold signed-energy condition that still makes (5) coercive.

References

- [1] P. Ivanisvili, R. Vershynin, and X. Xie, *Jackson’s inequality on the hypercube*, arXiv:2410.19949v1, 2024.
- [2] A. Eskenazis and P. Ivanisvili, *Polynomial inequalities on the Hamming cube*, Probability Theory and Related Fields 178 (2020), 235–287; arXiv:1902.02406.
- [3] K. Domelevo, P. Durcik, V. Fragkiadaki, O. Klein, D. Oliveira e Silva, L. Slavikova, and B. Wrobel, *Dimension-free estimates for low degree functions on the Hamming cube*, Studia Mathematica 280 (2025), no. 2, 161–176; arXiv:2401.07699.